

| STAT248 Lecture 5 Simple Linear Regression (Definition)

| 1 Simple Linear Regression (Definition)

| 1.1 Definition: Simple Linear Regression

Simple linear regression 使用 covariate $x \in \mathbb{R}$ 来预测 $y \in \mathbb{R}$:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

其中,

- β_0 和 β_1 为 parameters
 - β_0 表示 intercept, 即 $x = 0$ 时 y 的值
 - β_1 表示 slope, 即 x 变化一个 unit 时, y 的变化量
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| 1.2 Time Series Regression

对于 time series, 通常有两种方法预测 observed data $y_1, \dots, y_n$:

- 将 **time** 作为 **covariate**: 例如预测 population over time
 - 将 **lagged version of y** 作为 **covariate**: 例如取 $x_i = y_{i-1}$
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| 1.3 Solving Linear Regression (OLS)

标准库如 `statsmodels` 使用 least squares method, 即对于 observed data pairs $(x_1, y_1), \dots, (x_n, y_n)$, 我们希望 minimize the error sum of squares:

$$Q = \sum_{t=1}^n (y_t - \hat{y}_t)^2 = \sum_{t=1}^n (y_t - \beta_0 - \beta_1 x_t)^2$$

对 β_0 求导并令其为 0, 有:

$$\begin{aligned} \frac{\partial Q}{\partial \beta_0} &= \sum_{t=1}^n 2(y_t - \beta_0 - \beta_1 x_t) = 0 \\ \Rightarrow 2n\bar{y} - 2n\beta_0 - 2n\beta_1\bar{x} &= 0 \\ \Rightarrow \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1\bar{x} \end{aligned}$$

对 β_1 求导并令其为 0, 有:

$$\begin{aligned} \frac{\partial Q}{\partial \beta_1} &= \sum_{t=1}^n 2x_t(y_t - \beta_0 - \beta_1 x_t) = 0 \\ \Rightarrow 2n\overline{(xy)} - 2n\beta_0\bar{x} - 2n\beta_1\overline{(x^2)} &= 0 \\ \Rightarrow 2n\overline{(xy)} - 2n(\bar{y} - \beta_1\bar{x})\bar{x} - 2n\beta_1\overline{(x^2)} &= 0 \\ \Rightarrow \hat{\beta}_1 &= \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{(x^2)} - (\bar{x})^2} = \frac{\sum_{t=1}^n (y_t - \bar{y})(x_t - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2} \end{aligned}$$

综上:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} \quad \hat{\beta}_1 = \frac{\sum_{t=1}^n (y_t - \bar{y})(x_t - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2}$$

其中, $\bar{y} = \frac{y_1 + \dots + y_n}{n}$, $\bar{x} = \frac{x_1 + \dots + x_n}{n}$

1.4 Solving Linear Regression (MLE)

假设 errors 服从 Normal distribution, 即

$$\varepsilon_1, \dots, \varepsilon_n \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$$

则此时可以将 linear regression model 写作:

$$y_i \stackrel{\text{independent}}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$$

因此有 likelihood:

$$\begin{aligned} f_{y_1, \dots, y_n | \beta_0, \beta_1, \sigma}(y_1, \dots, y_n) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \\ \Rightarrow \text{log-likelihood} &= \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi}\sigma} - \sum_{i=1}^n \frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \\ &= -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \end{aligned}$$

注意到 sum of squared error 为 $Q = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 =: S(\beta_0, \beta_1)$, 对参数求导, 有:

$$\begin{aligned} \frac{\partial}{\partial \beta_0} \text{log-likelihood} &= -\frac{1}{2\sigma^2} \frac{\partial}{\partial \beta_0} S(\beta_0, \beta_1) = 0 \quad \Rightarrow \quad \frac{\partial}{\partial \beta_0} S(\beta_0, \beta_1) = 0 \\ \frac{\partial}{\partial \beta_1} \text{log-likelihood} &= -\frac{1}{2\sigma^2} \frac{\partial}{\partial \beta_1} S(\beta_0, \beta_1) = 0 \quad \Rightarrow \quad \frac{\partial}{\partial \beta_1} S(\beta_0, \beta_1) = 0 \\ \frac{\partial}{\partial \sigma} \text{log-likelihood} &= -\frac{n}{\sigma} + \frac{S(\beta_0, \beta_1)}{\sigma^3} = 0 \quad \Rightarrow \quad \sigma = \sqrt{\frac{S(\beta_0, \beta_1)}{n}} \end{aligned}$$

由此可以得到与之前相同的 $\hat{\beta}_0, \hat{\beta}_1$, 此外还可以得到:

$$\hat{\sigma}_{\text{MLE}} = \sqrt{\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{n}}$$

但事实上这是一个 biased estimator, unbiased estimator 为:

$$\hat{\sigma}_{\text{unbiased}}^2 = \frac{S(\hat{\beta}_0, \hat{\beta}_1)}{n-2}$$